Lecture (6)

Nile University

Analytical Geometry & Calculus I

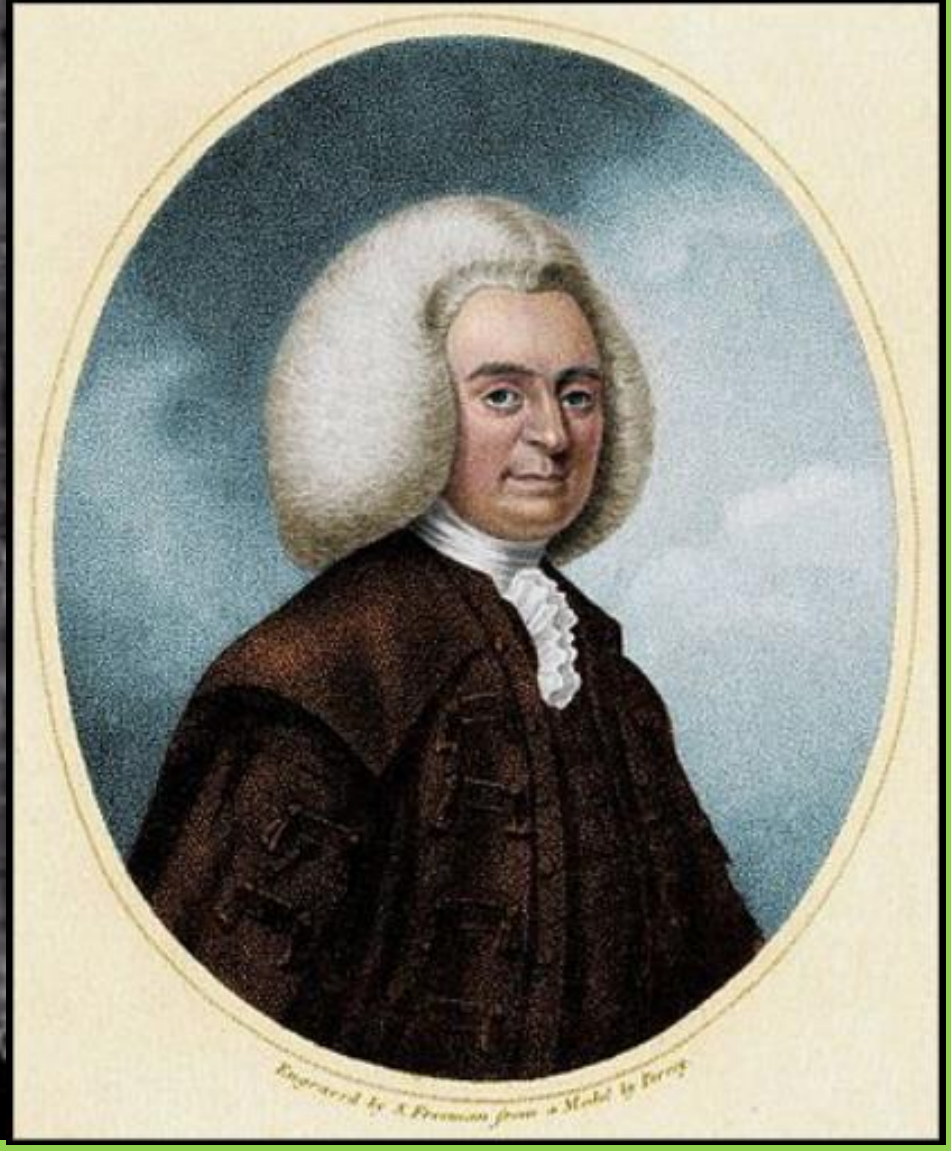
MTH 111

Outline

- **Taylor and Maclaurin series**
- **Analytic Geometry**
  - **Conic Sections: Circle**



Brook Taylor
(1685-1731)



Colin Maclaurin
(1698–1746)

Taylor and Maclaurin series

How to represent the function in power series(polynomials)?

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

?

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

MACLAURIN SERIES

- If a function *f(x)* has derivatives of all orders at x = 0, then the series

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \frac{f'''(0)}{3!}(x)^3 + \ldots$$

is called the **Maclaurin series for f(x)** .

Proof.. later

MACLAURIN SERIES

Example: Find the Maclaurin series of the function $f(x) = e^x$

$$f(x) = e^x \qquad f(0) = 1$$

$$f'(x) = e^x \qquad f'(0) = 1$$

$$f''(x) = e^x \qquad f''(0) = 1$$

$$f^{(n)}(x) = e^x \qquad f^{(n)}(0) = 1$$

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \frac{f'''(0)}{3!}(x)^3 + \ldots$$

MACLAURIN SERIES

Example: Find the Maclaurin series of the function $f(x) = e^x$

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \frac{f'''(0)}{3!}(x)^3 + \ldots$$

$$e^x = 1 + (x) + \frac{1}{2!}(x)^2 + \frac{1}{3!}(x)^3 + \ldots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

MACLAURIN SERIES

Guidelines for Finding Maclaurin Series

1. Differentiate f(x) several times

$$f(0),\ f''(0),\ f'''(0),\ldots,\ f^{(n)}(0)$$

Evaluate each derivative at $x=0$

2. Find the Maclaurin coefficients $\quad C_n = \dfrac{f^{(n)}(0)}{n!}$

MACLAURIN SERIES

Example: Find the Maclaurin series of the function $f(x) = \cos(x)$

$$f(x) = \cos(x) \qquad f(0) = 1$$

$$f'(x) = -\sin x \qquad f'(0) = 0$$

$$f''(x) = -\cos x \qquad f''(0) = -1$$

$$f'''(x) = \sin x \qquad f'''(0) = 0$$

$$f^{(4)}(x) = \cos x \qquad f^{(4)}(0) = 1$$

repetition:

MACLAURIN SERIES

Example: Find the Maclaurin series of the function $f(x) = \cos(x)$

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \frac{f'''(0)}{3!}(x)^3 + \ldots$$

$$\cos x = 1 + 0(x) + \frac{-1}{2!}(x)^2 + 0(x)^3 + \frac{1}{4!}(x)^4 \ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \ldots$$

MACLAURIN SERIES

Similarly you can proof the following.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \ldots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} \ldots$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} \ldots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \ldots$$

Maclaurin Series for Composite Function

- What about when f(x) = cos(x²)?

The series for cos x is

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$$

Now substitute by x² (replace each x by x²)

$$\cos x^2 = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \ldots$$

MACLAURIN SERIES

Example: Find the first three non zero terms of the Maclaurin series of the function $f(x) = x^2 e^{x^2}$

First find the Maclaurin series of e^x *(as before)*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

Replace each x by x^2

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \ldots$$

Multiply by x^2

$$x^2 e^{x^2} = x^2 + x^4 + \frac{x^6}{2!} + \frac{x^8}{3!} + \ldots$$

MACLAURIN SERIES

Examples (try): Find the first three non zero terms of the Maclaurin series of the following

$$f(x) = x\cosh\sqrt{x}$$

$$f(x) = x^3 \sinh(x^2)$$

$$f(x) = \ln(1 - x^2)$$

Taylor Series(more general)

- If a function *f(x)* has derivatives of all orders at x = c, then the series

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is called the Taylor series for f(x) at c.

- If c = 0, the series is the Maclaurin series for f(x) .

Taylor Series(more general)

Example: Find the Taylor series of the function

$$f(x) = \sin x \quad at \quad x = \frac{\pi}{2}$$

$$f(x) = \sin(x) \qquad f(\frac{\pi}{2}) = 1$$

$$f'(x) = \cos x \qquad f'(\frac{\pi}{2}) = 0$$

$$f''(x) = -\sin x \qquad f''(\frac{\pi}{2}) = -1$$

$$f'''(x) = -\cos x \qquad f'''(\frac{\pi}{2}) = 0$$

$$f^{(4)}(x) = \sin x \qquad f^{(4)}(\frac{\pi}{2}) = 1$$

repetition:

Taylor Series(more general)

Example: Find the Taylor series of the function

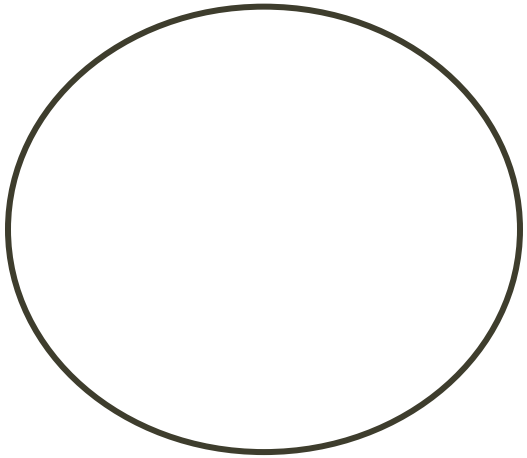
$$f(x) = \sin x \quad at \quad x = \frac{\pi}{2}$$

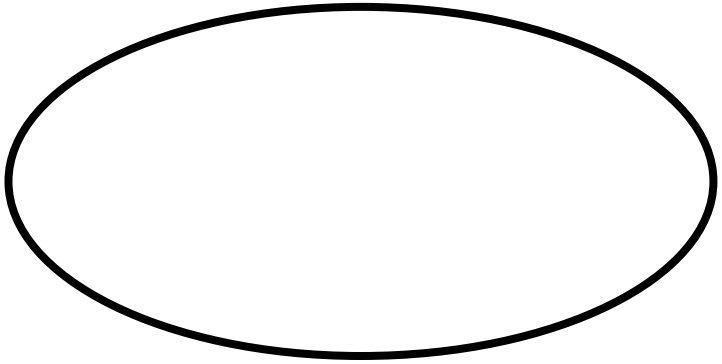
$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots$$

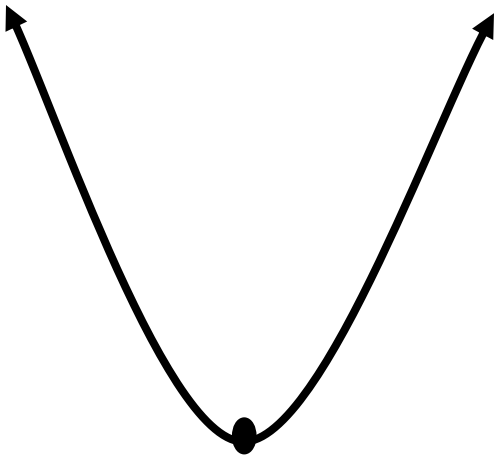
$$\sin x = 1 + 0(x - \frac{\pi}{2}) + \frac{-1}{2!}(x - \frac{\pi}{2})^2 + \frac{0}{3!}(x - \frac{\pi}{2})^3 + \frac{1}{4!}(x - \frac{\pi}{2})^4 + \ldots$$

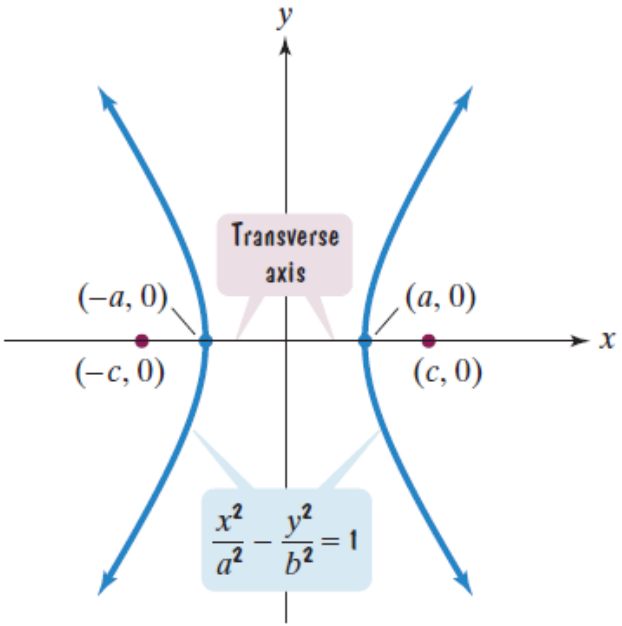
$$\sin x = 1 - \frac{1}{2!}(x - \frac{\pi}{2})^2 + \frac{1}{4!}(x - \frac{\pi}{2})^4 + \ldots$$

Lecture 6: part2 : Conic Sections







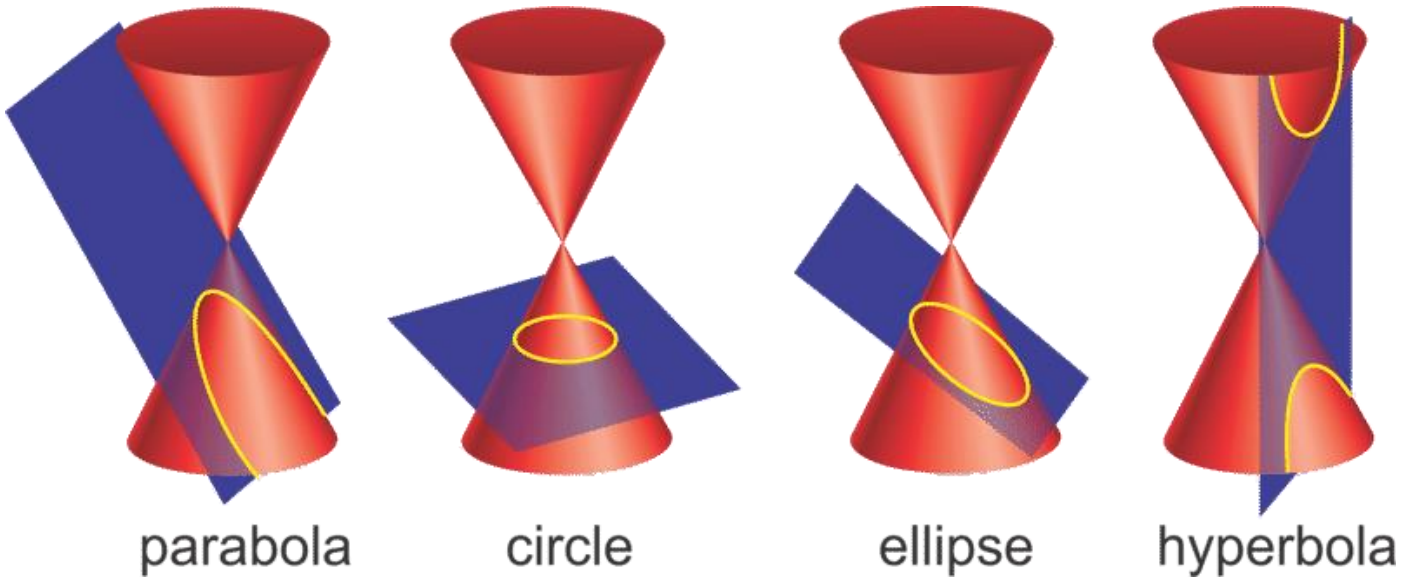


What are conic sections



Conic Sections

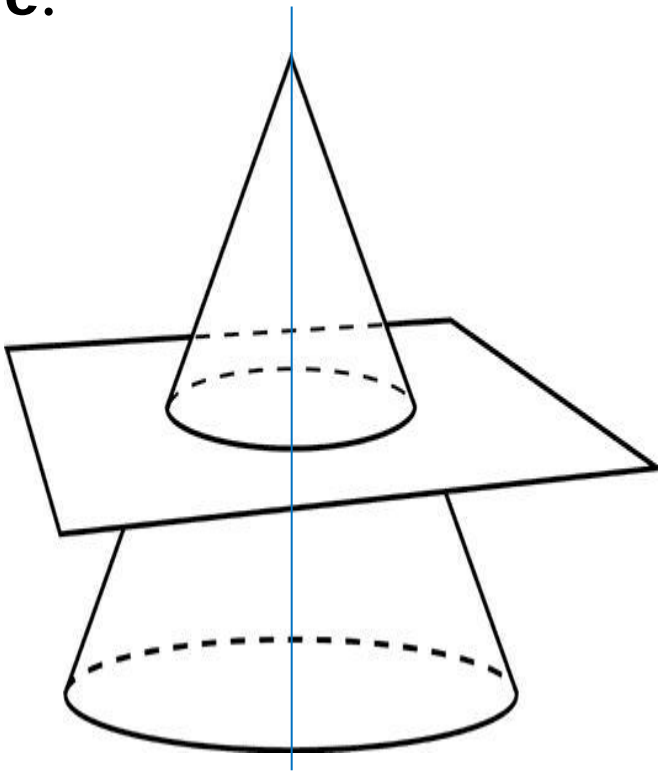
A conic section is a curve obtained as the intersection of a double cone with a plane.

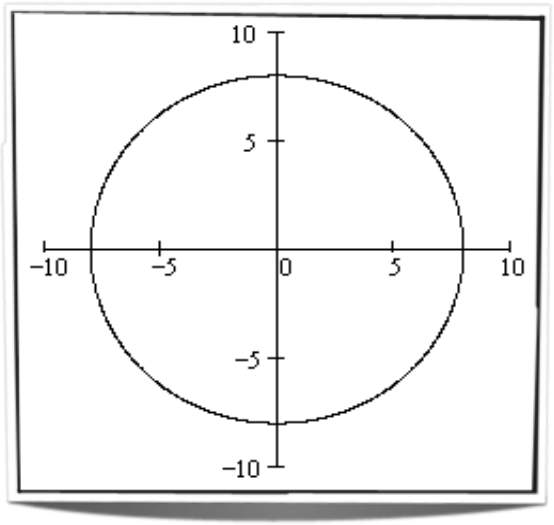


We can use Algebra to describe the equations and graphs of these shapes.

Circle

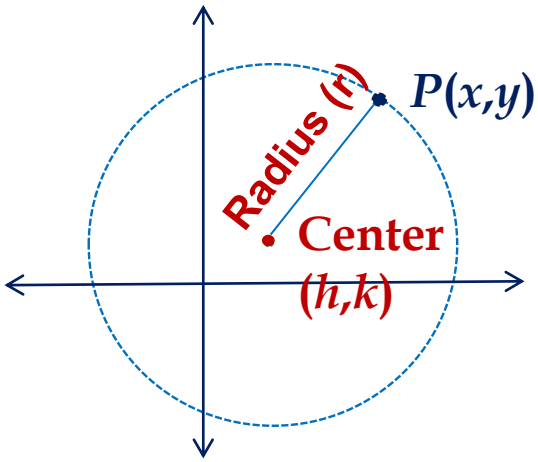
The plane that intersects the cone perpendicular to the axis of symmetry of the cone results in the curve of the circle.





Deriving the Equation of a Circle

Definition **A circle is a set of points in the plane that are equidistant from a fixed point in the plane (the center).**



Substitute with the values

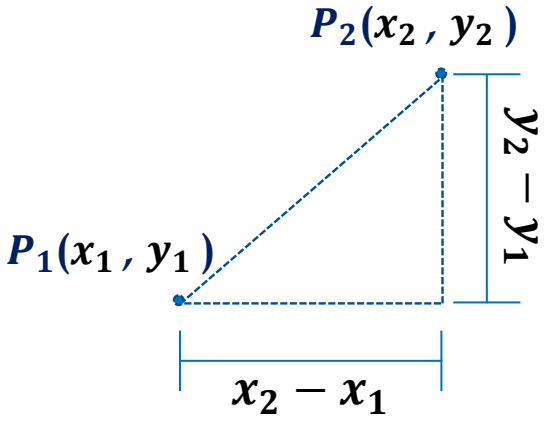
$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

Square both sides to get

$$r^2 = (x-h)^2 + (y-k)^2$$ ← **Center-radius Form**

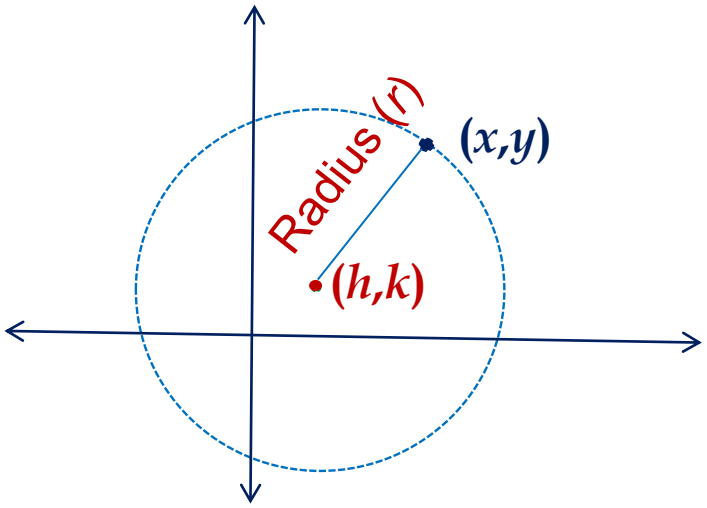
The distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



The Circle: Center-radius form of the equation

$$(x-h)^2+(y-k)^2=r^2$$



$(x+2)^2+(y+1)^2=9$ ⇒ **This is the equation of the circle of center (-2,-1) and radius =3.**

$(x+2)^2+(y+1)^2=0$ ⇒ **This is the equation of the point** (-2,-1)

$(x+1)^2+(y-3)^2=-1$ ⇒ **No graph**

The Circle: Example

Example **Find the center and radius for each of the following circles:**

- $\boldsymbol{(x-3)^2+(y+1)^2=16}$

  center $(3,-1), r=4$

- $\boldsymbol{(x+5)^2+(y-2)^2=15}$

  center $(-5,2), r=\sqrt{15}$

- $\boldsymbol{x^2+(y-2)^2-9=0}$

  center $(0,2), r=3$

The Circle: Example

Example **Write the equation of a circle centered at $(2, -7)$ and having a radius of 5.**

$$(x-2)^2 + (y+7)^2 = 25$$

Example **Write the equation of a circle with center $(2, 4)$ and passes through the point $(3, 7)$.**

The radius is the distance between $(2,4)$ and $(3,7)$.

$$r = \sqrt{(3-2)^2 + (7-4)^2} = \sqrt{10}$$

Circle equation

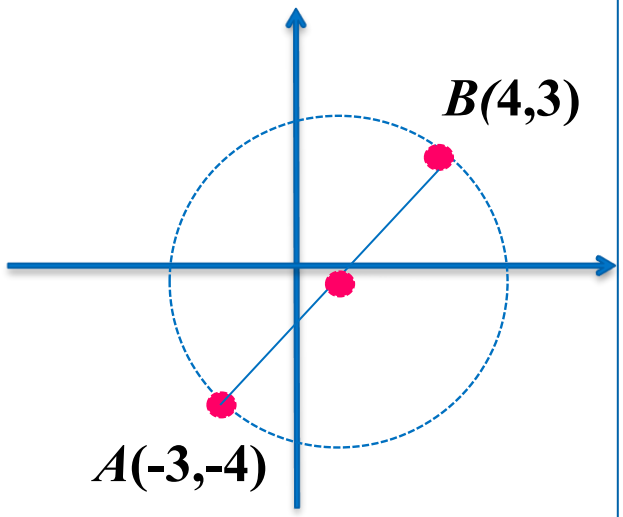
$$(x-2)^2 + (y-4)^2 = 10$$

The Circle: Example

Example Write the equation of a circle whose diameter is the line segment joining $A(-3,-4)$ and $B(4,3)$.

The center is the midpoint of the segment

$$\left(\frac{-3+4}{2}, \frac{-4+3}{2}\right) = \left(\frac{1}{2}, -\frac{1}{2}\right)$$



The radius is the distance from the center to a point on the circle. Use the distance formula:

$$r = \sqrt{(4-0.5)^2 + (3+0.5)^2} = \frac{7}{\sqrt{2}}$$

The equation of the circle $(x-0.5)^2 + (y+0.5)^2 = \frac{49}{2}$

The Circle: Completing the square

Consider the equation

$$x^2 \pm ax + b = \left(x \pm \frac{a}{2}\right)^2 - \left(\frac{a}{2}\right)^2 + \mathrm{b}$$

Example Write the equation of the following circle in a center-radius form.

$$\boldsymbol{x^2 + y^2 + 6x - 4y = 12}$$

$\boldsymbol{x^2 + 6x + y^2 - 4y = 12}$ **(Complete square)**

$\boldsymbol{(x+3)^2 - 9 + (y-2)^2 - 4 = 12}$

$\boldsymbol{\rightarrow (x+3)^2 + (y-2)^2 = 25}$

The Circle: Equation of tangent

Example **Find the equation of the tangent and the normal lines of the circle, $(x-1)^2+(y+2)^2=25$, at the point $(4,2)$**

First check that the point is on the circle

$\text{LHS}=(4-1)^2+(2+2)^2=25=\text{RHS}$

To get the tangent

$$2(x-1)+2(y+2)y'=0$$

$$y'=\frac{1-x}{y+2}$$

$$y'\Big|_{(4,2)}=m=\frac{1-4}{2+2}=-\frac{3}{4}$$

Equation of tangent

$$\frac{y-2}{x-4}=-\frac{3}{4}\rightarrow y=-\frac{3}{4}x+5$$

Equation of normal

$$\frac{y-2}{x-4}=\frac{4}{3}\rightarrow y=\frac{4}{3}x-\frac{10}{3}$$



See you next week

MACLAURIN SERIES(proof)

- We start by supposing that *f* is any function that can be represented by a power series

$$f(x) = c_0 + c_1(x) + c_2(x)^2 + c_3(x)^3 + c_4(x)^4 + \ldots$$

- substitute x=0:

$$f(0) = c_0$$

- By differentiate the series and substitute x=0:

$$f'(x) = c_1 + 2c_2(x) + 3c_3(x)^2 + 4c_4(x)^3 + \ldots$$

$$f'(0) = c_1$$

MACLAURIN SERIES(proof)

- Differentiate again and substitute x=0:

$$f''(x) = 2c_2 + 2 \cdot 3c_3(x) + 3 \cdot 4c_4(x)^2 + \ldots$$

$$f''(0) = 2C_2 \qquad \Rightarrow C_2 = \frac{f''(0)}{2}$$

- Let's apply the procedure one more time.

$$f'''(x) = 2 \cdot 3c_3 + 2 \cdot 3 \cdot 4c_4(x) + 3 \cdot 4 \cdot 5c_5(x)^2 \ldots$$

$$C_3 = \frac{f'''(0)}{(3)(2)} = \frac{f'''(0)}{(3)!}$$

MACLAURIN SERIES(proof)

If we continue to differentiate and substitute $x = 0$, we obtain:

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \frac{f'''(0)}{3!}(x)^3 + \ldots$$

Taylor Series(more general)

Examples(try): Find the Taylor series of the following functions

i. $f(x) = e^x$, at $x = 1$

ii. $f(x) = \ln(1 + x)$ at $x = 1$